

## Homework 5 (Graded)

Professor: Dr. Mudassir Shabbir

Deadline: 11:59AM Apr 5, 2026.

**Instructions.** This is a graded homework. Submit your solutions as a single PDF. Show all work; correct answers without justification will receive no credit. You may collaborate on high-level ideas but must write up solutions independently.

1. [25 points] *This is an interval DP over substrings.*

Given a string  $S = s_1 s_2 \cdots s_n$ , let  $\text{LPS}(i, j)$  denote the length of the longest palindromic subsequence in substring  $S[i..j]$ .

Use the string:

$$S = \text{BANANAS}$$

(indexed from 1 to 7).

- (a) [5 pts] Give a precise definition of the state  $\text{LPS}(i, j)$  and all base cases.
- (b) [8 pts] Write the recurrence for the two cases  $s_i = s_j$  and  $s_i \neq s_j$ .
- (c) [7 pts] Compute enough table entries to determine  $\text{LPS}(1, 7)$ . Show your fill order by increasing substring length.
- (d) [3 pts] Report the final length of the LPS of BANANAS.
- (e) [2 pts] Give asymptotic time and space complexity of the bottom-up DP.

2. [25 points] *You are given an array and must partition it into exactly  $k$  contiguous segments to minimize total cost.*

Let  $A[1..n]$  be an array of positive integers. For any segment  $A[t..i]$ , define its segment cost as

$$\text{cost}(t, i) = \left( \sum_{r=t}^i A[r] \right)^2.$$

You must partition  $A[1..n]$  into exactly  $k$  nonempty contiguous segments.

For this problem use:

$$A = [3, 1, 4, 1, 5, 9], \quad n = 6, \quad k = 3.$$

- (a) [5 pts] Define a DP state  $\text{DP}[i][j]$  that represents an optimization over prefixes and segment counts.
- (b) [5 pts] Give base cases for  $\text{DP}[i][1]$  and any impossible states.
- (c) [7 pts] Derive a recurrence for  $\text{DP}[i][j]$  by choosing the last segment boundary.
- (d) [5 pts] Compute the optimal value for  $\text{DP}[6][3]$ . Show intermediate values clearly for all states with  $j \in \{1, 2, 3\}$ .
- (e) [3 pts] State time complexity of the straightforward bottom-up implementation in terms of  $n$  and  $k$ , and explain how prefix sums change the cost of evaluating each transition.

3. [25 points] *Determine if one string can be formed by interleaving two others.*

You are given three strings:

$$s_1 = \text{aabcc}, \quad s_2 = \text{dbbca}, \quad s_3 = \text{aadbcbcac}.$$

- (a) [5 pts] Give a precise state definition and all base cases.
- (b) [8 pts] Derive the recurrence. (Hint: consider where the last character of  $s_3[i+j]$  came from.)
- (c) [6 pts] Fill enough of the DP table to determine  $\text{Inter}[5][5]$  (whether  $s_3[1..10]$  is an interleaving of  $s_1$  and  $s_2$ ).

- (d) **[3 pts]** Report whether  $s_3$  can be formed as an interleaving of  $s_1$  and  $s_2$ .
- (e) **[3 pts]** Give asymptotic running time and space complexity.

4. **[25 points]** *Find the path from top-left to bottom-right that minimizes total cost.*

You are given a  $4 \times 5$  grid with positive integer costs. You may only move right or down.

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 3 | 1 | 5 | 1 |
| 2 | 2 | 4 | 1 | 5 |
| 5 | 0 | 2 | 3 | 0 |
| 6 | 1 | 1 | 2 | 1 |

- (a) **[5 pts]** Give the state definition and all base cases.
- (b) **[8 pts]** Derive the recurrence relation for  $DP[i][j]$ , considering that we can reach  $(i, j)$  from either  $(i - 1, j)$  or  $(i, j - 1)$ .
- (c) **[6 pts]** Fill the complete DP table for the given grid.
- (d) **[3 pts]** Report the minimum path sum to reach  $(4, 5)$ .
- (e) **[3 pts]** State the time and space complexity, and describe how to reconstruct one optimal path.